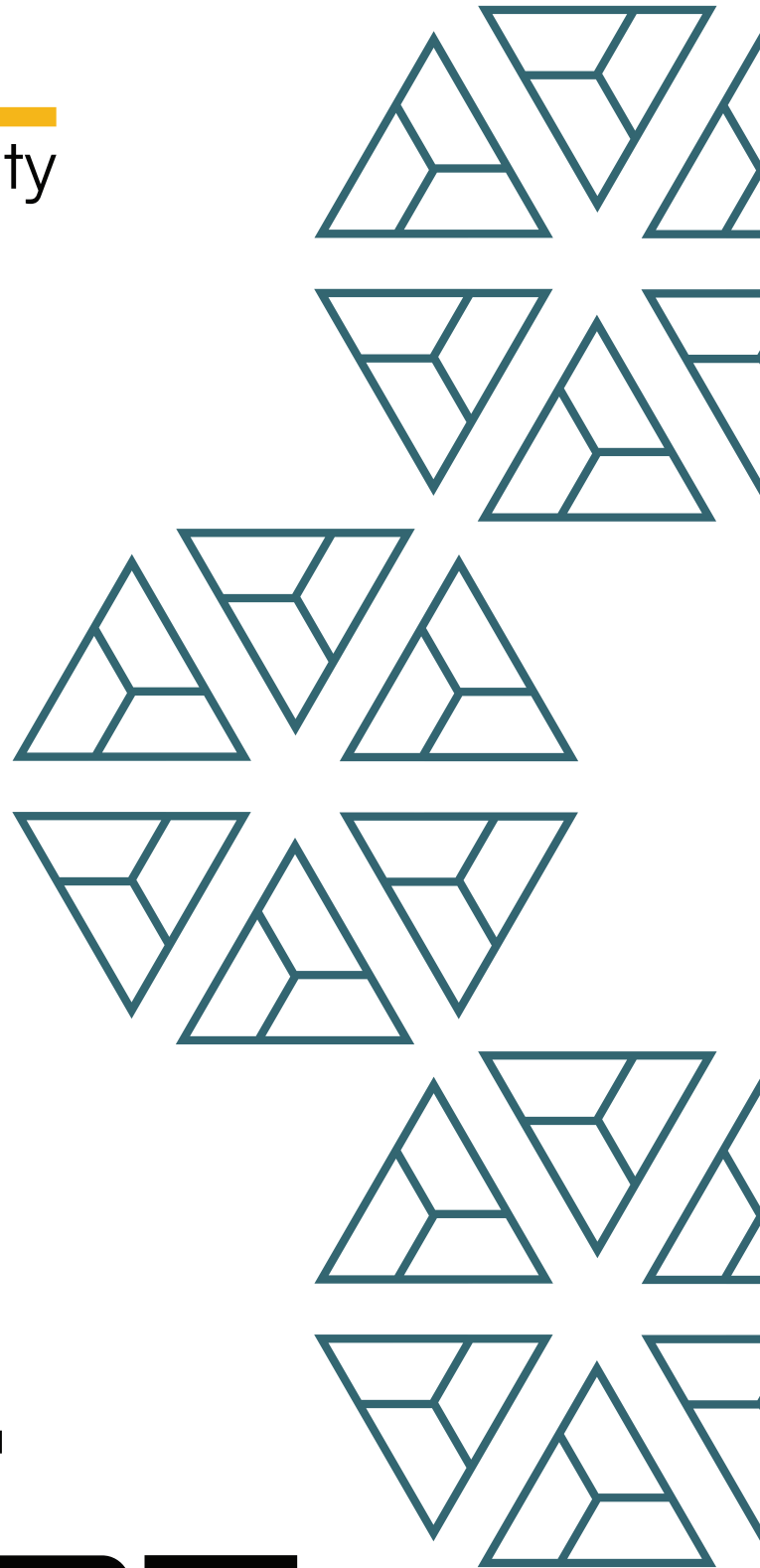




BAIL
security



Lista Dao
V3 Differential

FINAL REPORT

May '2026

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Security assessment projects are time-boxed and often reliant on information that may be provided by a client, its affiliates, or its partners. As a result, the findings documented in this report should not be considered a comprehensive list of security issues, flaws, or defects in the target system or codebase.

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By engaging in a smart contract audit, the contract owner acknowledges and agrees to the terms of this disclaimer.

1. Project Details

Important:

Please ensure that the deployed contract matches the source-code of the last commit hash.

Project	Lista Dao – V3 Differential - Audit Report
Website	lista.org
Language	Solidity
Methods	Manual Analysis
Github repository	https://github.com/lista-dao/lista-v3/tree/d577c43aa68229908ca8337faea1f0974e0bf2e5
Resolution 1	https://github.com/lista-dao/lista-v3/tree/9227bf0db9f66dfb216c7d6cb5fc3a9316690090

2. Detection Overview

Severity	Found	Resolved	Partially Resolved	Acknowledged (no changes made)	Failed resolution	Open
High	1	1				
Medium						
Low	1	1				
Informational	4	2		2		
Governance						
Total	6	4		2		

2.1 Detection Definitions

Severity	Description
High	The problem poses a significant threat to the confidentiality of a considerable number of users' sensitive data. It also has the potential to cause severe damage to the client's reputation or result in substantial financial losses for both the client and the affected users.
Medium	While medium level vulnerabilities may not be easy to exploit, they can still have a major impact on the execution of a smart contract. For instance, they may allow public access to critical functions, which could lead to serious consequences.
Low	Poses a very low-level risk to the project or users. Nevertheless the issue should be fixed immediately
Informational	Effects are small and do not post an immediate danger to the project or users
Governance	Governance privileges which can directly result in a loss of funds or other potential undesired behavior

3. Detection

Bailsec has been tasked with a differential review of the following scope:

<https://github.com/lista-dao/lista-v3/tree/d577c43aa68229908ca8337faea1f0974e0bf2e5/src>

/core

/periphery

The base repositories are from standard UniswapV3:

<https://github.com/Uniswap/v3-core/tree/main/contracts>

<https://github.com/Uniswap/v3-periphery/tree/main/contracts>

The differentials are determined as follows:

NEW:

- periphery/base/ERC721PermitUpgradeable.sol: <https://www.diffchecker.com/fn1gqV6t/>

MODIFIED:

- core/NoDelegateCall.sol: <https://diffchecker.com/GXrUsH2d>
- core/UniswapV3Factory.sol: <https://diffchecker.com/Z9F43XdM>
- core/UniswapV3Pool.sol: <https://diffchecker.com/Hqdoo67q>
- core/UniswapV3PoolDeployer.sol: <https://diffchecker.com/AM7caYzp>
- core/libraries/Oracle.sol: <https://diffchecker.com/DbxI8lzm>
- core/libraries/Position.sol: <https://diffchecker.com/9cv7ivBN>
- core/libraries/SqrtPriceMath.sol: <https://diffchecker.com/HwqOaAtc>
- core/libraries/SwapMath.sol: <https://diffchecker.com/C6yLZ4f2>
- core/libraries/Tick.sol: <https://diffchecker.com/wYUknslh>
- core/libraries/TickBitmap.sol: <https://diffchecker.com/oriPuouQ>
- periphery/NonfungiblePositionManager.sol: <https://diffchecker.com/5LW3Xqcj>
- periphery/NonfungibleTokenPositionDescriptor.sol: <https://diffchecker.com/GBzNzPVo>
- periphery/SwapRouter.sol: <https://diffchecker.com/oNWg4Oud>
- periphery/V3Migrator.sol: <https://diffchecker.com/T0Jv8KvZ>
- periphery/base/ERC721Permit.sol: <https://diffchecker.com/TSxBTa9W>
- periphery/base/LiquidityManagement.sol: <https://diffchecker.com/9b7cCOoK>
- periphery/base/PeripheryPayments.sol: <https://diffchecker.com/SaQjweXb>
- periphery/base/PeripheryPaymentsWithFee.sol: <https://diffchecker.com/1zOFPnP1>
- periphery/base/PoolInitializer.sol: <https://diffchecker.com/Ro8ELOBR>
- periphery/base/SelfPermit.sol: <https://diffchecker.com/KDMAmaCA>
- periphery/lens/Quoter.sol: <https://diffchecker.com/5LoTxP63periphery/lens/QuoterV2.sol>
<https://diffchecker.com/ehAtAW45>
- periphery/lens/TickLens.sol: <https://diffchecker.com/l6brN5vB>
- periphery/lens/UniswapInterfaceMulticall.sol: <https://diffchecker.com/dR7NkQmF>
- periphery/libraries/CallbackValidation.sol: <https://diffchecker.com/8NLjZaXJ>
- periphery/libraries/LiquidityAmounts.sol: <https://diffchecker.com/zJPWhbYS>
- periphery/libraries/NFTDescriptor.sol: <https://diffchecker.com/6H4Whze9>
- periphery/libraries/NFTSVG.sol: <https://diffchecker.com/nttWTEXI>
- periphery/libraries/OracleLibrary.sol: <https://diffchecker.com/RNYoeODv>
- periphery/libraries/PoolAddress.sol: <https://diffchecker.com/36AsxRtj>
- periphery/libraries/PoolTicksCounter.sol: <https://diffchecker.com/HcuB77FH>
- periphery/libraries/PositionValue.sol: <https://diffchecker.com/9rY5vLAY>
- periphery/libraries/SqrtPriceMathPartial.sol: <https://diffchecker.com/ZoXWKNvX>
- periphery/libraries/TransferHelper.sol: <https://diffchecker.com/huo8040Q>

IDENTICAL:

- core/libraries/BitMath.sol: No changes
- core/libraries/FixedPoint128.sol: No changes
- core/libraries/FixedPoint96.sol: No changes
- core/libraries/FullMath.sol: No changes
- core/libraries/LiquidityMath.sol: No changes
- core/libraries/LowGasSafeMath.sol: No changes
- core/libraries/SafeCast.sol: No changes
- core/libraries/TickMath.sol: No changes
- core/libraries/TransferHelper.sol: No changes
- core/libraries/UnsafeMath.sol: No changes
- periphery/base/BlockTimestamp.sol: No changes
- periphery/base/Multicall.sol: No changes
- periphery/base/PeripheryImmutableState.sol: No changes
- periphery/base/PeripheryValidation.sol: No changes
- periphery/libraries/BytesLib.sol: No changes
- periphery/libraries/ChainId.sol: No changes
- periphery/libraries/HexStrings.sol: No changes
- periphery/libraries/Path.sol: No changes
- periphery/libraries/PositionKey.sol: No changes
- periphery/libraries/TokenRatioSortOrder.sol: No changes

Due to the sake of simplicity, changes in the import paths are not separately listed as changes. It is required that imports from node_modules match the correct versions and this is not covered during this audit.

Global

Issue_01	Several interfaces still point to UniswapV3
Severity	Informational
Description	The following interfaces still point to UniswapV3 comments: - INonfungiblePositionManager - IPeripheryImmutableState - ISwapRouter Furthermore, some interfaces are internally named still with UniswapV3, especially those callback related ones However, we do not recommend changing this, as changing the callback relations will break compatibility. Furthermore, most market makers and integrators assume these callbacks to indeed be named after UniswapV3, which means their architecture still works under the current ListaV3 code.
Recommendations	Consider adjusting the comments for these interfaces. We explicitly do not recommend adjusting import paths or contract namings as this could result in breaking imports and/or callback compatibility.
Comments / Resolution	Acknowledged.

NonfungiblePositionManager

Changes:

- Renaming from UniswapV3 to Lista
- License change
- Upgrade support
- Inherits ERC721PermitUpgradeable instead of ERC721Permit

<https://www.diffchecker.com/5LW3Xqcj/>

Issue_02	Gap is redundant for non-inherited contracts
Severity	Informational
Description	Currently, the NFPM contract exposes a gap: <pre><i>uint256[45] private __gap;</i></pre> A gap is generally helpful in contracts that are meant to be inherited and used as implementation contracts for proxies. This is however not true for the NFPM, as this contract is not meant to be inherited and as such, it does not benefit to add a gap.
Recommendations	Consider removing the gap if it is indeed clear that this contract will never be inherited.
Comments / Resolution	Resolved.

Issue_03	Initialization-skip can produce dangerous side-effects
Severity	Informational
Description	With most contracts, it is obvious that initialization is required for a contract to work, as it often sets important addresses or default settings. With the NFPM, this is not totally clear as only the <code>_tokenDescriptor</code>, <code>_nextId</code> and <code>_nextPoolId</code> are set. Intuitively, one could think the contract can work without it as well, which appears indeed to be true. However, if a mint happens and <code>initialize</code> is called afterwards, it will reset the <code>_nextId</code> and <code>_nextPoolId</code> values to 1, which can then produce unexpected side-effects.
Recommendations	Consider ensuring that the initialization happens at the same time as proxy deployment.
Comments / Resolution	Acknowledged.

ERC721PermitUpgradeable

This contract is based on Uniswap's ERC721Permit contract and supports upgradeability.

Changes:

- Upgradeability support

<https://www.diffchecker.com/fn1gqV6t/>

Issue_04	Missing gap for inherited contracts
Severity	Low
Description	<code>ERC721PermitUpgradeable</code> is a custom upgradeable parent inherited by <code>NonfungiblePositionManager</code> and introduces proxy storage variables <code>_nameHash</code> and <code>_versionHash</code>, but does not reserve a storage gap after them. Since this parent is linearized before NFPM's own storage, any future storage variable added to <code>ERC721PermitUpgradeable</code> would shift the storage slots of NFPM's child variables, including <code>_poolIds</code>, <code>_poolIdToPoolKey</code>, <code>_positions</code>, <code>_nextId</code>, <code>_nextPoolId</code>, and <code>_tokenDescriptor</code>. This can corrupt or orphan existing position data after an upgrade, especially because mappings depend on their root storage slot. NFPM's own <code>__gap</code> does not mitigate this issue, as it only protects storage additions at the NFPM level and not additions in inherited parent contracts. If the contract is not yet deployed, a dedicated gap such as <code>uint256[48] private __gap;</code> should be added to <code>ERC721PermitUpgradeable</code>; if already deployed, the parent layout should be treated as frozen and future state should be added only through safe namespaced/unstructured storage patterns.
Recommendations	Consider adding a gap to this contract.
Comments / Resolution	Resolved.

NoDelegateCall

Changes:

- License change

<https://www.diffchecker.com/GXrUsH2d/>

ListaV3Factory

Changes:

- Renaming from UniswapV3 to Lista
- License change

<https://www.diffchecker.com/Z9F43XdM/>

No issues found in changes.

ListaV3Pool

Changes:

- Renaming from UniswapV3 to Lista
- License change

<https://www.diffchecker.com/Hqdoo67q/>

No issues found in changes.

ListaV3PoolDeployer

Changes:

- Renaming from UniswapV3 to Lista
- License change

<https://www.diffchecker.com/AM7caYzp/>

No issues found in changes.

Oracle

Changes:

- License change

<https://www.diffchecker.com/Dbxl8lzm/>

No issues found in changes.

Position

Changes:

- License change

<https://www.diffchecker.com/9cv7ivBN/>

No issues found in changes.

SqrtPriceMath

Changes:

- License change

<https://www.diffchecker.com/HwqOaAtc/>

No issues found in changes.

SwapMath

Changes:

- License change

<https://www.diffchecker.com/C6yLZ4f2/>

No issues found in changes.

Tick

Changes:

- License change

<https://www.diffchecker.com/wYUknslh/>

No issues found in changes.

TickBitMap

Changes:

- License change

<https://www.diffchecker.com/oriPuouQ/>

No issues found in changes.

NonfungibleTokenPositionDescriptor

Changes:

- Renaming from UniswapV3 to Lista

<https://www.diffchecker.com/GBzNzPVo/>

No issues found in changes.

SwapRouter

Changes:

- Renaming from UniswapV3 to Lista

<https://www.diffchecker.com/oNWg4Oud/>

No issues found in changes.

V3Migrator

Changes:

- Renaming from UniswapV3 to Lista

<https://www.diffchecker.com/T0Jv8KvZ/>

No issues found in changes.

ERC721Permit

Changes:

- no changes

<https://www.diffchecker.com/T0Jv8KvZ/>

No issues found in changes.

LiquidityManagement

Changes:

- Renaming from UniswapV3 to Lista

<https://diffchecker.com/9b7cC0oK>

No issues found in changes.

PeripheryPayments

Changes:

- no changes

<https://www.diffchecker.com/SaOjweXb/>

No issues found in changes.

PeripheryPaymentsWithFee

Changes:

- no changes

www.diffchecker.com/1zOFPnP1/

No issues found in changes.

PoolInitializer

Changes:

- Renaming from UniswapV3 to Lista

<https://www.diffchecker.com/Ro8ELOBR/>

No issues found in changes.

SelfPermit

Changes:

- no changes

<https://www.diffchecker.com/KDMAmaCA/>

No issues found in changes.

Quoter

Changes:

- Renaming from UniswapV3 to Lista

<https://www.diffchecker.com/5LoTxP63/>

No issues found in changes.

QuoterV2

Changes:

- Renaming from UniswapV3 to Lista

<https://www.diffchecker.com/ehAtAW45/>

No issues found in changes.

TickLens

Changes:

- Renaming from UniswapV3 to Lista

<https://www.diffchecker.com/I6brN5vB/>

No issues found in changes.

ListaInterfaceMulticall

Changes:

- Renaming from UniswapV3 to Lista

<https://www.diffchecker.com/dR7NkQmF/>

No issues found in changes.

CallbackValidation

Changes:

- Renaming from UniswapV3 to Lista

<https://www.diffchecker.com/8NLjZaXJ/>

No issues found in changes.

LiquidityAmounts

Changes:

- Renaming from UniswapV3 to Lista

<https://www.diffchecker.com/zJPWhbYS/>

No issues found in changes.

NFTDescriptor

Changes:

- Renaming from UniswapV3 to Lista

<https://www.diffchecker.com/6H4Whze9/>

No issues found in changes.

NFTSVG

Changes:

- Renaming from UniswapV3 to Lista

<https://www.diffchecker.com/nttWTEXI/>

Issue_05	Incorrect sig in generateSVG natspec
Severity	Informational
Description	Currently, the NATSPEC is as follows: <pre> /* address: "0xe8ab59d3bcde16a29912de83a90eb39628cfc163", msg: "Forged in SVG for Lista in 2021 by 0xe8ab59d3bcde16a29912de83a90eb39628cfc163", sig: "0x2df0e99d9cbfec33a705d83f75666d98b22dea7c1af412c584f7d6 26d83f02875993df740dc87563b9c73378f8462426da572d7989de8 8079a382ad96c57b68d1b", version: "2" */ </pre> This is however incorrect, as NFTSVG keeps the original Uniswap SVG provenance signature while modifying the signed message from "Uniswap" to "Lista". Since the signature is bound to the exact message bytes, the retained signature no longer proves that 0xe8ab...c163 signed the displayed Lista statement. This does not affect runtime behavior because the block is only a Solidity comment, but it creates incorrect provenance/attribution metadata and should either be re-signed, reverted to the original message, or removed.
Recommendations	Consider acknowledging this issue or reverting to the previous signed message.
Comments / Resolution	Resolved.

OracleLibrary

Changes:

- Renaming from UniswapV3 to Lista

<https://www.diffchecker.com/RNYoeODv/>

No issues found in changes.

PoolAddress

Changes:

- Renaming from UniswapV3 to Lista
- changing POOL_INIT_CODE_HASH

<https://www.diffchecker.com/36AsxRtj/>

Issue_06	Incorrect <code>POOL_INIT_CODE_HASH</code>
Severity	High
Description	<code>PoolAddress.POOL_INIT_CODE_HASH</code> is hardcoded and used by the periphery to deterministically derive Lista V3 pool addresses. This value is consumed by <code>PoolAddress.computeAddress()</code> and therefore affects <code>SwapRouter</code>, <code>LiquidityManagement</code>, <code>CallbackValidation</code>, <code>Quoter</code>, <code>QuoterV2</code>, <code>TickLens</code>, and the NFT position descriptor. The configured hash matches <code>keccak256(type[ListaV3Pool].creationCode)</code> only when <code>ListaV3Pool</code> is compiled with optimizer runs 50. However, the repository default Foundry profile uses optimizer runs 500, which produces a different pool creation-code hash. Unless the production deployment explicitly compiles <code>ListaV3Pool</code> with the same settings used to derive the hardcoded hash, the factory-created pool address will differ from the address computed by the periphery. As a result, the periphery can resolve the wrong pool address and callback validation will reject legitimate callbacks from actual deployed pools. This can make core periphery flows unusable, including swaps, minting/increasing liquidity through NFPM, quoting, lens reads, and metadata generation.
Recommendations	Consider either adjusting the <code>POOL_INIT_CODE_HASH</code> or sticking to 50 runs. As a safety measure, consider verifying the actual deployed init code hash after the deployment before adding any pool. And if it mismatches, redeploy everything with the correct hash.
Comments / Resolution	Resolved.

PoolTicksCounter

Changes:

- Renaming from UniswapV3 to Lista

<https://diffchecker.com/HcuB77FH>

No issues found in changes.

PositionValue

Changes:

- Renaming from UniswapV3 to Lista

<https://www.diffchecker.com/9rY5vLAY/>

No issues found in changes.

SqrtPriceMathPartial

Changes:

- Renaming from UniswapV3 to Lista

<https://www.diffchecker.com/ZoXWKNvX/>

No issues found in changes.

TransferHelper

Changes:

- none

<https://www.diffchecker.com/huo8040Q/>

No issues found in changes.